Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	27 October 2026
Team ID	LMP-EV-2026
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Connect to the EV market CSV in Tableau Desktop. Use the available vehicle, model, manufacturer, model year, category, battery, range, efficiency, price, annual sales, customer rating, market share, country, charging-station and CO2-saved fields where present.	High
2	Data Processing	Prepare the connected training CSV by removing null values and duplicate records and correcting data types. Configure model_year, category, battery_type, manufacturer and manufacturing_country as dimensions and the relevant technical/commercial measures as numeric fields. The test CSV was supplied but was not connected to the workbook.	High
3	Business Analysis	Executive KPIs: SUM(annual_sales_units), COUNTD(vehicle_id), AVG(range_km), AVG(price_usd) in USD, and AVG(customer_rating). Analyze sales by category/manufacturer; battery-type share; MEDIAN(range_km) by battery type; COUNTD(vehicle_id) by country and battery type; price versus range; and rating/battery-capacity comparisons by category.	High
4	Business Insights & Analytical Outputs	Provide nine completed worksheets: Executive Summary; Category Sales; Manufacturer Share (Treemap); Battery Performance; Battery Tech Share; Global Manufacturing Map; Price vs Range; Rating vs Battery; Country vs Battery Matrix. Combine them in the EV Market Performance & Strategy Dashboard. Only Story Point 1, Global EV Market & OEM Overview, is completed; Story Points 2-4 are planned future work.	High
5	User Interaction & Analysis	Provide the global filters specified in the project brief: model_year, category, battery_type and manufacturing_country. Support chart selection, tooltips, comparisons and Use as Filter actions where configured. The public website provides access to the embedded dashboard and a direct Tableau Public link.	High
6	Data Update & Reporting	When a new dataset version is available, the project owner manually replaces or reconnects the CSV in Tableau Desktop, checks the worksheets and dashboard, republishes to Tableau Public and verifies the GitHub Pages embed. The current implementation is a published snapshot; no automated/scheduled refresh is evidenced.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Use clearly titled worksheets and dashboard sections, readable axes, legends, data labels, tooltips and filters. The website provides navigation to the embedded dashboard and a direct Tableau Public link. Confirm mobile readability in final testing.
NFR-2	Security	The dashboard and website are public, read-only views. Do not publish confidential or personal data. Edit permissions for the Tableau workbook and GitHub repository remain with the project owner. The public site does not provide private login/authentication.
NFR-3	Reliability	Use the expected CSV schema and verify key fields and aggregations after source changes. Check worksheets, filters, dashboard actions, story navigation and the Tableau Public embed before sharing an updated version. A direct Tableau Public link is the fallback if the iframe fails.
NFR-4	Performance	Keep the HTML/CSS landing page lightweight. Tableau rendering depends on workbook size, data volume, Tableau Public and network conditions. No fixed response-time target is claimed because no performance test result was supplied.
NFR-5	Availability	The website uses GitHub Pages and the interactive visualization uses Tableau Public. Access depends on both external platforms and internet connectivity; no independent uptime guarantee or offline mode is implemented.
NFR-6	Scalability	The workbook can be extended if added records and categories retain compatible fields. Larger data volumes may require extract and workbook optimization. No separate warehouse or automatic scaling layer is implemented.

Implementation boundary

These requirements describe the evidenced CSV + Tableau + static website implementation. Automated ingestion, scheduled refresh, database storage, authentication, API services, Flask and Streamlit are not claimed as implemented.

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Confirmed data preparation and scope: nulls and duplicate records were removed and data types corrected. The training CSV is connected; the test CSV was supplied but not connected. The price_usd measure is USD ($). Only Tableau Story Point 1 is completed; Story Points 2-4 remain planned.